AquaSentry Home: Personal In-Tank Water Quality & Arsenic Sentry

Dedicated Household Storage Tank Surveillance Node



1. Concept: How It Works for One Person / Household

Unlike the autonomous lake catamaran which roams open waters, **AquaSentry Home** is engineered specifically for **one person or single family**.

- **Placement**: Installed directly inside your domestic overhead water storage tank (e.g. Sintex 500L / 1000L / 2000L black/blue tank) or underground groundwater sump. It can either **float on a 3D-printed buoyant collar** or **clamp to the inspection rim**.
- **Private Reporting**: The unit connects directly to your home Wi-Fi (2.4 GHz). It reports real-time water data straight to your personal smartphone, laptop, or home dashboard without needing any paid subscription.
- **Local Alerts**:
- An onboard 360° RGB LED ring glows through a clear dome on the tank lid:
  - 🟢 **Solid Green**: Water Safe (Arsenic $< 10\ \mu g/L$, TDS normal, pH 6.8–7.8).
  - 🟡 **Amber**: Caution (Exceeds WHO guideline, filter maintenance advised).
  - 🔴 **Pulsing Red + Audio Beep**: Hazardous contamination ($> 50\ \mu g/L$ or acute turbidity spike).
- Pushes instant WhatsApp / Telegram / local Web notifications to your phone.

2. Motherboard Options with Wi-Fi

Motherboard Option	Price Range (INR)	Wi-Fi / Specs	Verdict for Personal Tank Use
ESP32 DevKit V1 (30 / 38 Pin) *(Recommended)*	**₹350 – ₹550**	802.11 b/g/n (2.4 GHz) + BLE, Dual-Core 240MHz, 520KB SRAM	**Best Choice.** Ubiquitous in India (Robu.in, ElectronicsComp, Amazon). Enough horsepower to run on-edge neural model, Wi-Fi web server, and all analog pins.
Raspberry Pi Pico W	**₹550 – ₹750**	802.11n Wi-Fi, Dual-Core ARM Cortex-M0+ 133MHz	Great for MicroPython lovers, but has only 3 ADC channels and slightly higher analog noise.
ESP32-S3 DevKit	**₹650 – ₹950**	Dual-Core LX7 with Vector AI acceleration, Wi-Fi 4 + BLE 5	High-end overkill; great if you want advanced on-chip machine learning or camera integration.
NodeMCU ESP8266	**₹220 – ₹320**	802.11 b/g/n, Single-Core 80MHz	Too limited: only 1 single analog pin (A0); requires external multiplexer to read multiple probes.

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3. Complete Bill of Materials (BOM) in Indian Rupees (₹ / INR)

All parts are standard, widely stocked robotics/electronics components available across India from platforms like **Robu.in**, **ElectronicsComp.com**, **Quartzcomponents.com**, or **Amazon.in**.

A. Microcontroller & Wi-Fi Core

Component	Function on Device	Price Range (INR)
ESP32 DevKit V1 (30-Pin Micro-USB / Type-C)	Motherboard brain: reads all sensors, controls the sampling pump, connects to home Wi-Fi, and hosts local user dashboard.	**₹350 – ₹550**
ADS1115 16-Bit 4-Channel I2C ADC Module	Precision analog-to-digital converter ($7.8\ \mu V$ resolution) to read micro-volt/nano-amp electrochemical stripping peaks.	**₹220 – ₹380**
LM358 Dual Op-Amp (or MCP6002 Rail-to-Rail) + basic resistors & capacitors on mini perfboard	Miniature potentiostat circuit: provides voltage sweep (-0.8V to +0.4V) and Transimpedance Amplifier (TIA) for current measurement.	**₹80 – ₹150**

B. Electrochemical Arsenic Sensing Unit

Component	Function on Device	Price Range (INR)
Screen-Printed 3-Electrode Strip *(MAS-G1 / DropSens Gold Working Electrode)*	**[The One Specialized Part]** Stripping voltammetry strip (Gold WE, Ag/AgCl RE, Carbon CE) to detect dissolved Arsenic (As^{3+}).	**₹350 – ₹650** *(per strip / cartridge)*
12V / 5V DC Micro Peristaltic Dosing Pump	Draws a 10–15 mL fresh water sample into an enclosed 3D-printed cell once a day, then purges it. **Keeps the sensor strip dry between tests, extending strip life from days to months!**	**₹450 – ₹850**
Single-Channel 5V Relay Module or MOSFET Driver (IRF520)	Allows the 3.3V ESP32 to switch the 12V peristaltic pump on and off.	**₹40 – ₹90**
Food-Grade Silicone Tubing (2mm ID / 4mm OD, 2 meters)	Suction hose dipping 30 cm into the tank water to pull fresh water without surface dust.	**₹80 – ₹150**

C. Continuous Auxiliary Water Quality Probes

Component	Function on Device	Price Range (INR)
DS18B20 Waterproof Stainless Probe	Reads water temperature. Temperature dictates arsenic stripping kinetic rates and feeds calibration.	**₹120 – ₹220**
Analog TDS / Electrical Conductivity Sensor Module	Continuously measures Total Dissolved Solids / salinity. Instantly detects if municipal water is mixed with tanker/borewell water.	**₹280 – ₹450**
Analog Turbidity Sensor Module (Optical IR)	Detects suspended sediment/rust inside the tank; notifies user when tank needs manual cleaning.	**₹450 – ₹750**
Analog pH Sensor Kit (E-201-C Glass Probe + 4502C Board) *(Optional)*	Measures acidity/alkalinity (pH $5.5 - 9.5$). Arsenic toxicity/speciation is pH-dependent.	**₹950 – ₹1,500**

D. Power, Alerts & Enclosure

Component	Function on Device	Price Range (INR)
12V 2A Regulated DC Power Adapter + DC Jack Extension	Powers the unit from the nearest household plug point (e.g. geyser socket or rooftop water pump switchboard).	**₹250 – ₹450**
LM2596 DC-DC Step-Down Buck Converter	Steps 12V down to clean 5.0V to power ESP32, ADC, and sensors.	**₹60 – ₹120**
WS2812B NeoPixel RGB LED Ring (8 or 12 LEDs)	360-degree color ring on the tank top lid giving instant visual potability status (Green / Amber / Red).	**₹120 – ₹220**
Active 5V Piezo Buzzer	High-pitch local audible alarm if acute arsenic ($> 50\ \mu g/L$) or dangerous contamination occurs.	**₹20 – ₹40**
3D-Printed PETG Enclosure & Flange (approx. 250g)	Watertight cylindrical shell, internal sample well, and tank mounting bracket.	**₹250 – ₹450** *(material cost)*
O-Ring Gasket / Silicone Sealant + Stainless Screws	Hermetically seals electronics against high tank humidity and condensation.	**₹80 – ₹150**

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4. Total Cost Summary in Rupees (INR)

- **Minimalist Budget DIY Build** *(ESP32 + Arsenic Potentiostat + TDS + Temp + Pump + 3D Shell + Power Adapter)*:
  ₹2,350 – ₹3,400
- **Complete Comprehensive Sentry** *(ESP32 + Arsenic ASV + pH + Turbidity + TDS + Temp + Pump + LED Ring + Buzzer + 3D Shell)*:
  ₹4,200 – ₹6,100

(Compare this to commercial industrial water analyzers which retail for ₹45,000 to ₹1,50,000+!)

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5. System Wiring & Communication Architecture



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6. Daily Operation Cycle (How It Reports to You)

1. **Continuous Baseline Monitoring (24/7)**:

2. The TDS, pH, and Temperature probes stay submerged in the tank. Every 15 minutes, the ESP32 takes a reading.
3. If tank TDS spikes (e.g. municipal water switched to unpurified groundwater), you receive an immediate notification on your phone.
4. **Automated Morning Arsenic Stripping Cycle (e.g., 6:00 AM)**:
5. Before the household begins morning cooking/bathing, the ESP32 activates the peristaltic pump for 6 seconds.
6. 15 mL of water is drawn into the internal sealed chamber.
7. Anodic Stripping Voltammetry (ASV) is performed: deposition at $-0.8\ \mathrm{V}$ followed by an anodic stripping ramp to $+0.4\ \mathrm{V}$.
8. The `AquaNet-V1` neural model computes risk:
    - **If Safe**: NeoPixel glows serene Green. App updates: *"Arsenic: 0.007 mg/L — Water is safe for domestic use."*
    - **If Unsafe**: NeoPixel turns flashing Red, piezo buzzer sounds, and your phone receives an urgent alert: *"High Arsenic Detected (0.058 mg/L)! Do not drink or cook before filtration."*
9. The pump reverses, purging the 15 mL sample back out, keeping the sensor strip dry until the next test.
10. **Local Private Dashboard**:
11. Type `http://aquasentry.local` (or the ESP32's local IP address like `192.168.1.145`) on any phone or laptop connected to your home Wi-Fi to view historical trend charts, battery/power health, and filter lifespan.

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7. 3D Printing Recommendations for the Personal Tank Shell

- **Filament**: **PETG** (Food-grade certified, zero BPA, handles temperatures up to $75^{\circ}\mathrm{C}$ on hot summer rooftops).
- **Layer Height**: **0.20 mm** with **4 perimeters** for 100% watertight sealing.
- **Mounting Design**:
- Print a curved hanging bracket that hooks over the lip of standard 400mm / 500mm inspection manholes of Sintex or Supreme water tanks.
- The electronics dome remains above the waterline, while the modular probes suspend 30–50 cm into the center of the water column.
